MOTILAL NEHRU NATIONAL INSTITUTE OF TECHNOLOGY, ALLAHABAD

DATA STRUCTURE LAB

ASSIGNMENT- 7 (Stack & Queue)

1. The contents of a stack S are as follows:

   Stack(S) 99 20 4 27
   Index 0 1 2 3 4 5 6
   The Stack can store a maximum of seven elements and the top pointer currently points at index 3. Write a program to show the stack contents and indicate the position of the top pointer after each of the following stack operations:
   o Push (S, 5)
   o Push (S, 7)
   o Pop (S)
   o Pop (S)
   o Pop (S)

   o Deduce the contents of an empty Queue after the execution of the following operations in sequence:
   o Enqueue (6)
   o Enqueue (8)
   o Enqueue (–1)
   o Dequeue()
   o Enqueue (7)
   o Dequeue()
   o Dequeue()

2. Write a C program to evaluate the prefix expression using Stack.
   Input : -+7*45+20
   Output : 25

3. Write a C program to evaluate the postfix expression using Stack.
   Input: 5 6 7 + * 8 -
   Output: 57

4. Write a C program to evaluate the infix expression using Stack
   Input: 150 * (20 + 10) / 15
   Output: 300

5. Write a C program to convert the infix expression to prefix expression.
   Infix: (A + B) * (C + D)
   Prefix: * + A B + C D

6. Write a C program to convert the infix expression to postfix expression.
   Infix: (A + B) * (C + D)
   Postfix: A B + C D + *
7. Write a C program that accepts a string and reverse it using a Stack.
8. Write a program to implement two stacks in a single array.
9. Write a program to check the valid parenthesis in a given expression.
   Example: ((a-b/c)*(a/k-l))
10. Write a program for Tower of Hanoi problem using stack.

11. Write a program to implement Circular Queue.
    a. Insert an element into the circular queue.
    b. Delete an element from the circular queue.
    c. Display the contents of the circular queue.

12. Write a program to implement Priority Queue.
    a. Insert an element into the priority queue.
    b. Delete an element from the priority queue.
    c. Display the contents of the priority queue.

13. Write a C program to delete the nth element of a queue. Firstly, this code checks if the queue is empty or if the position to delete is invalid. If the position is valid, it deletes the first n-1 elements, then deletes the nth element by calling the dequeue() function.

    Expected Output:
    Insert some elements into the queue:
    Queue elements are: 1 2 3 4 5
    Delete the 7th element of the said queue:
    Error: Invalid position
    Delete the 3rd element of the said queue:
    Queue elements are: 4 5